Lesson 3 Teacher's Key

001 Game Creator (Windows only)

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution

One Input Trigger script, plus two Actor Trigger scripts on the bullet Actor itself:

```
Input Trigger: Spacebar (Pressed) ->
  Map – Spawn Actor(bullet)

bullet Actor: Automatic Movement -> Move = true, Direction = Up

bullet Actor Trigger: Left/Entered Camera (Left) ->
  Actor – Destroy/Revive Actor(this, Destroy)
```

Level 2 - Complete Worked Solution

Input Triggers set to "Pressed" already only fire once per physical press (not every frame held), so Level 1's fire check needs no changes for Level 2. An optional tunable cooldown, if a slower fire rate is wanted, uses a Timer/Spawn object's own interval instead of scripted variables (the more idiomatic 001 Game Creator approach):

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: play on fire

Import sfx_shoot.wav (Resources > Import > Sound), then add a "Play Sound" event directly after Map – Spawn Actor:

```
Input Trigger: Spacebar (Pressed) ->
  Map – Spawn Actor(bullet)
  Play Sound(sfx_shoot.wav)
```

Visual: give the bullet its own look

Set Tint/Opacity and Size directly on the bullet Actor Template's own Graphics Options and General settings - since these are Template-level settings (not scripted per-spawn), every copy Map – Spawn Actor creates automatically looks this way with zero extra events.

Structure: keep bullet behavior on the bullet Actor

main's Input Trigger script should only ever trigger a shot (Map – Spawn Actor + sound) - everything about what a bullet actually does once it exists (its Automatic Movement, its own Left/Entered Camera cleanup trigger, its own Template-level look) belongs on the bullet Actor

Template itself. Keeping this boundary clean is what let the Visual idea above need zero scripting at all.

Complete Level 3 solution (all 3 goals combined)

Everything above combined - the bullet Actor Template's own settings/triggers, plus main's fire script:

```
bullet Actor Template settings:
  Automatic Movement -> Move = true, Direction = Up
  Graphics Options -> Tint/Opacity set, Size increased

bullet Actor Trigger: Left/Entered Camera (Left) ->
  Actor – Destroy/Revive Actor(this, Destroy)

Input Trigger: Spacebar (Pressed) ->
  Map – Spawn Actor(bullet)
  Play Sound(sfx_shoot.wav)
```

Alternate Path Notes: Textual Scripting

Reiterate to students in person: this mode switch is genuinely irreversible per-project. Confirm they created a separate throwaway project before letting them proceed.

Exact text-command names should be checked against the installed version's Script Editor autocomplete or the wiki before class, since this course otherwise treats 001 Game Creator as visual-only and hasn't needed to track Textual syntax changes.

The Level 3 goals map onto Textual commands too - the sound idea is a typed equivalent of Play Sound, and the visual idea needs no scripting either way, since it's a Template-level setting in both modes.

- This path exists for students who've realized they prefer typing code (maybe from Godot or GameMaker) and want to see whether 001 Game Creator supports that style too - it does, with this one important caveat.
- Because of the irreversibility, this should stay an optional, teacher-supervised exploration rather than a path any student defaults into.